PGCP-AI

# PRACTICAL MACHINE LEARNING

## Day 2 : Data Pre-processing

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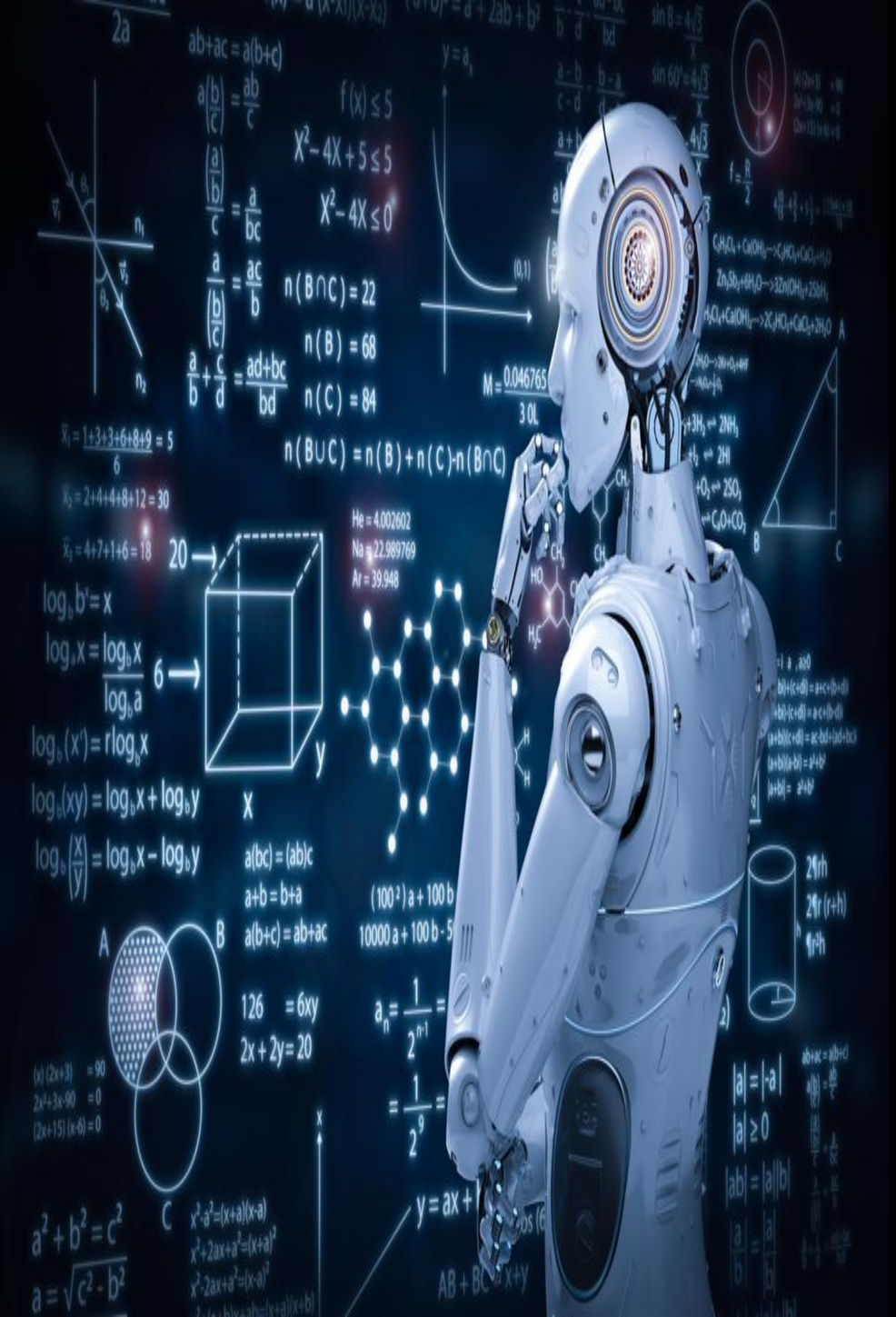


# Agenda

- Data
- Different types of Data
- Types of Attributes
- Pre-processing
- Transformations
- Feature scaling
- Visualization

# What is data?

- **Collection of data objects and their attributes**

- **An attribute is a property or characteristic of an object**
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as **variable, field, characteristic, or feature**

- **A collection of attributes describe an object**
  - Object is also known as **record, point, case, sample, entity, or instance**

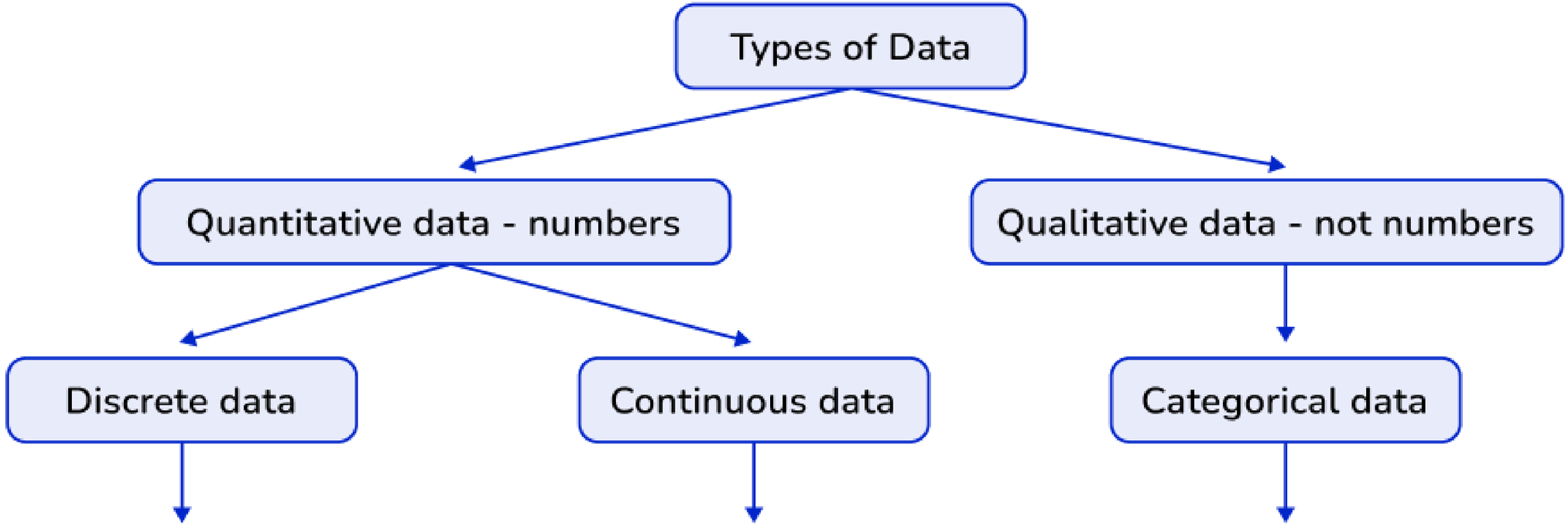
**Attributes**

**Objects**

| *Tid* | Refund | Marital Status | Taxable Income | Cheat |
|---|---|---|---|---|
| 1 | Yes | Single | 125K | No |
| 2 | No | Married | 100K | No |
| 3 | No | Single | 70K | No |
| 4 | Yes | Married | 120K | No |
| 5 | No | Divorced | 95K | Yes |
| 6 | No | Married | 60K | No |
| 7 | Yes | Divorced | 220K | No |
| 8 | No | Single | 85K | Yes |
| 9 | No | Married | 75K | No |
| 10 | No | Single | 90K | Yes |

# Types of Data

The different types of data we need to know are:




- Number of students
- Number of employees
- Score of a game

- Heights of people
- Daily temperature
- Time needed on a task

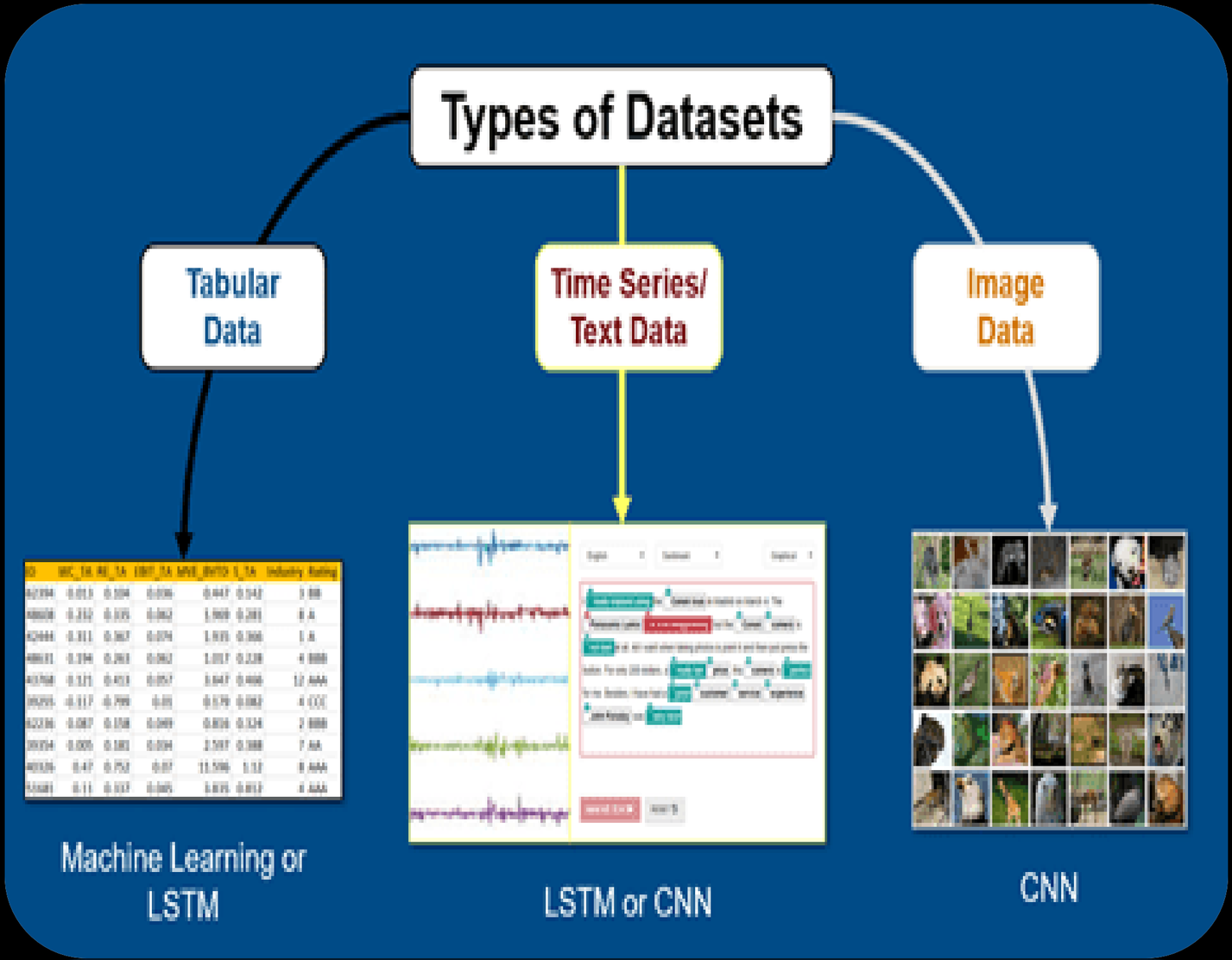
- Hair color
- Eye color
- Favorite color

# Types of attributes

- **There are different types of attributes**
  - Nominal
    - Examples: ID numbers, eye color, zip codes
  - Ordinal
    - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
  - Interval
    - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
  - Ratio
    - Examples: temperature in Kelvin, length, time, counts

# Types of data sets

- **Record**
  - Data matrix
  - Document data
  - Transaction data
- **Graph**
  - World Wide Web
  - Molecular structures
- **Ordered**
  - Spatial data
  - Temporal (time series) data
  - Sequential data
  - Genetic sequence data



# Record data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

| First Name | Last Name | Address | City | Age |
|---|---|---|---|---|
| Mickey | Mouse | 123 Fantasy Way | Anaheim | 73 |
| Bat | Man | 321 Cavern Ave | Gotham | 54 |
| Wonder | Woman | 987 Truth Way | Paradise | 39 |
| Donald | Duck | 555 Quack Street | Mallard | 65 |
| Bugs | Bunny | 567 Carrot Street | Rascal | 58 |
| Wiley | Coyote | 999 Acme Way | Canyon | 61 |
| Cat | Woman | 234 Purrfect Street | Hairball | 32 |
| Tweety | Bird | 543 | Itotltaw | 28 |

Records

# Data matrix

- **If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute.**

- **Such data set can be represented by an *m* x *n* matrix, where there are *m* rows, one for each object, and *n* columns, one for each attribute**

| Projection of x Load | Projection of y load | Distance | Load | Thickness |
|---|---|---|---|---|
| 10.23 | 5.27 | 15.22 | 2.7 | 1.2 |
| 12.65 | 6.25 | 16.22 | 2.2 | 1.1 |

# Document data

- **Each document becomes a 'term' vector,**
  - each term is a component (attribute) of the vector
  - the value of each component is the number of times the corresponding term occurs in the document.

| | team | coach | play | ball | score | game | win | lost | timeout | season |
|---|---|---|---|---|---|---|---|---|---|---|
| document 1 | 3 | 0 | 5 | 0 | 2 | 6 | 0 | 2 | 0 | 2 |
| document 2 | 0 | 7 | 0 | 2 | 1 | 0 | 0 | 3 | 0 | 0 |
| document 3 | 0 | 1 | 0 | 0 | 1 | 2 | 2 | 0 | 3 | 0 |

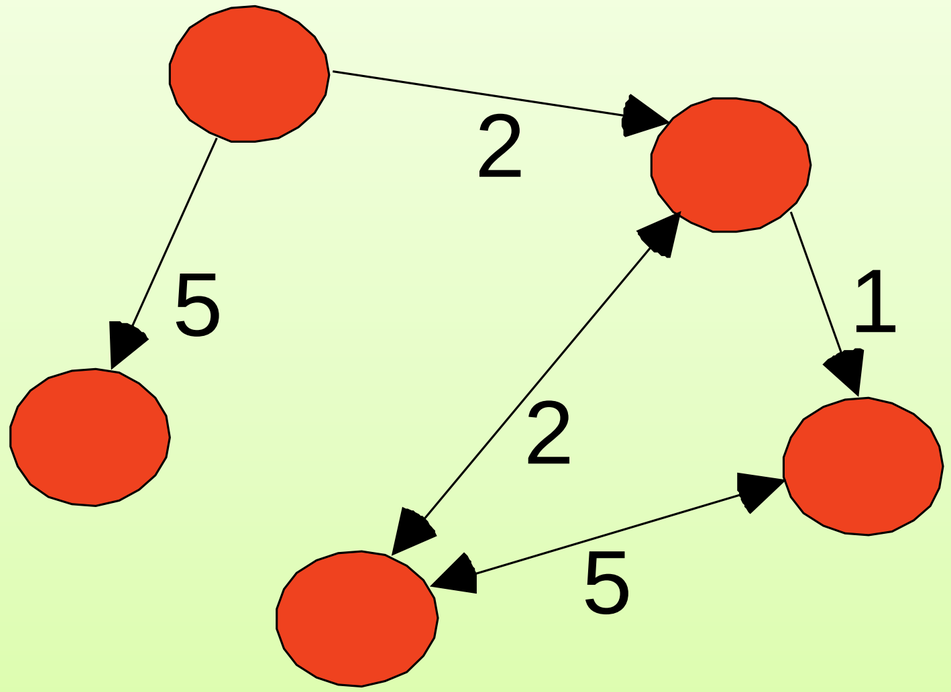
# Transaction data

- **A special type of record data, where**
  - Each record (transaction) involves a set of items.
  - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

| *TID* | *Items* |
|---|---|
| 1 | Bread, Coke, Milk |
| 2 | Beer, Bread |
| 3 | Beer, Coke, Diaper, Milk |
| 4 | Beer, Bread, Diaper, Milk |
| 5 | Coke, Diaper, Milk |

# Graph data

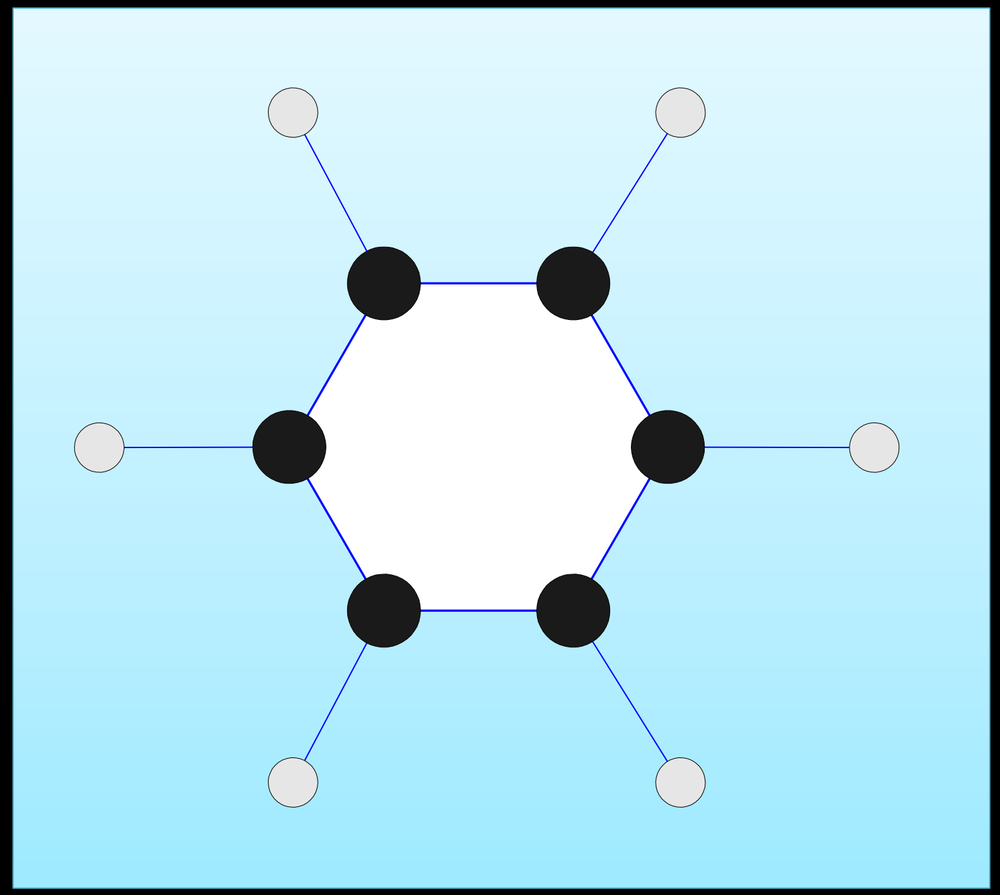
- **Examples: Generic graph and HTML Links**




```
<a href="papers/papers.html#bbbb">
Data Mining </a>
<li>
<a href="papers/papers.html#aaaa">
Graph Partitioning </a>
<li>
<a href="papers/papers.html#aaaa">
Parallel Solution of Sparse Linear System of Equations </a>
<li>
<a href="papers/papers.html#ffff">
N-Body Computation and Dense Linear System Solvers
```

# Chemical data

- **Benzene molecule: $C_6H_6$**



# Ordered data

- **Sequences of transactions**

Items/Events

( A B) (D) (C E)
( B D) (C) (E)
( C D) (B) (A E)

An element of the sequence

# Ordered data

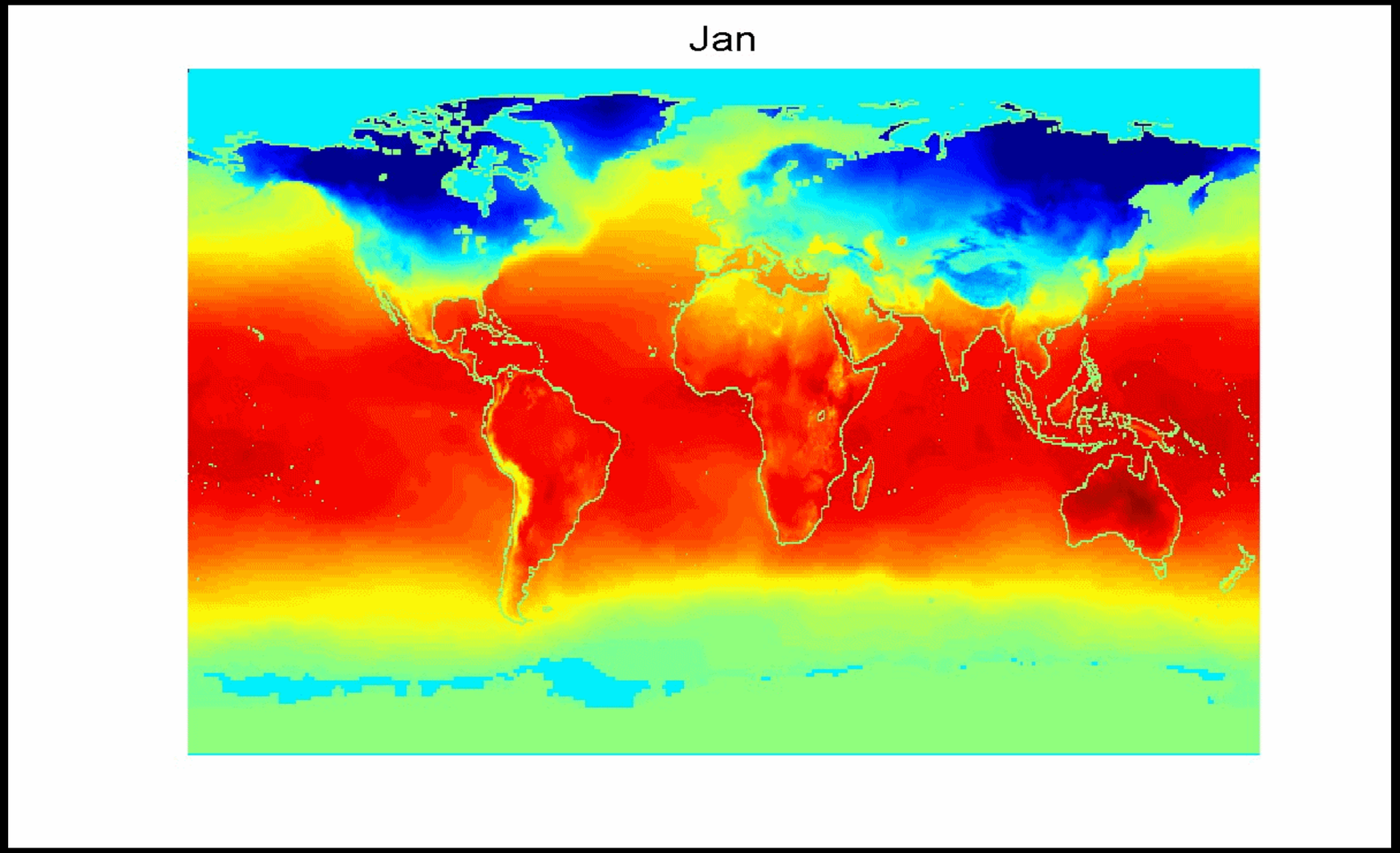
- **Genomic sequence data**

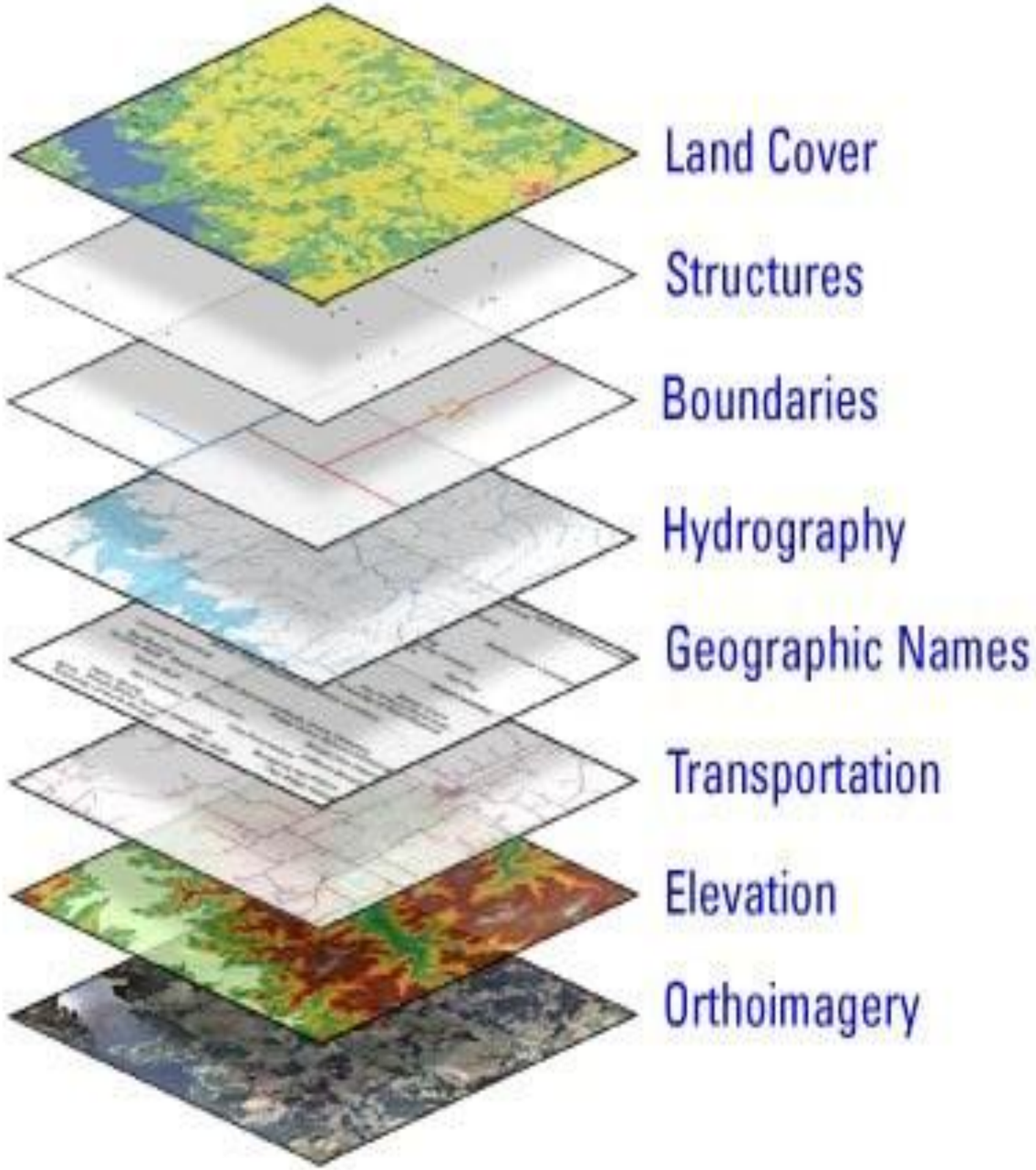
```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

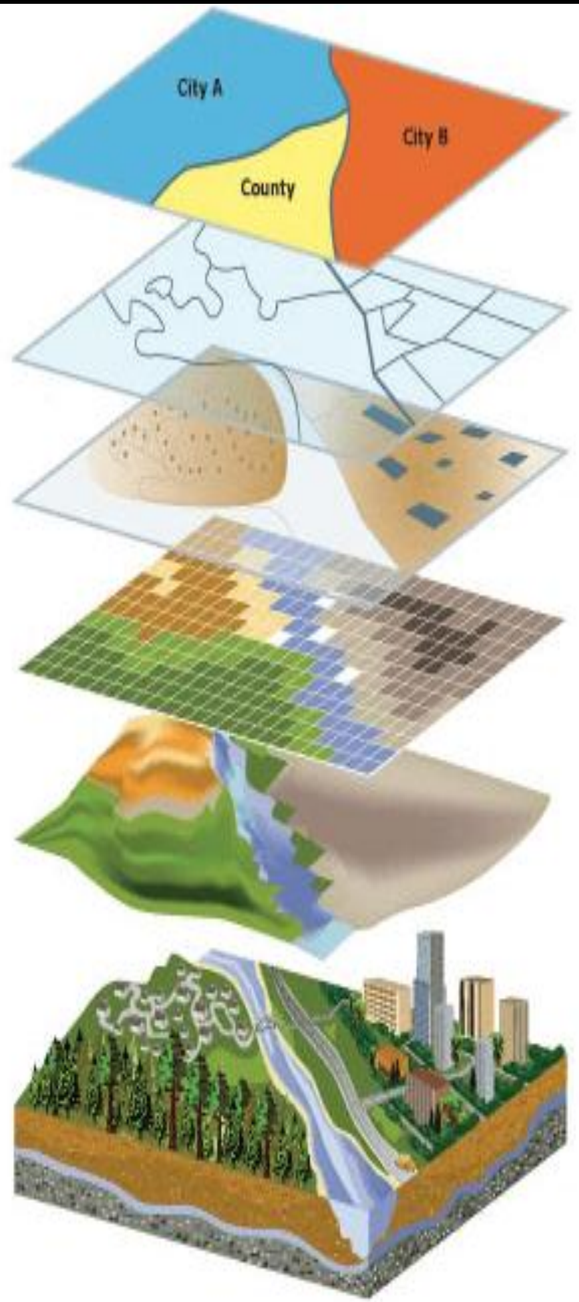
# Ordered data

- **Spatio-temporal data**

Average monthly temperature of land and ocean




Land Cover
Structures
Boundaries
Hydrography
Geographic Names
Transportation
Elevation
Orthoimagery

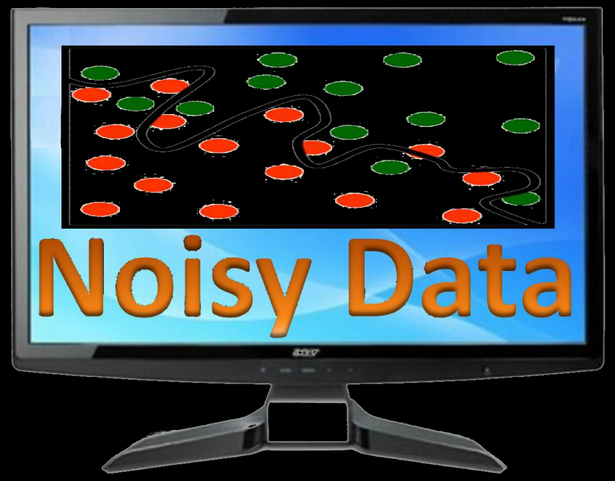

City A
City B
County

# Data quality

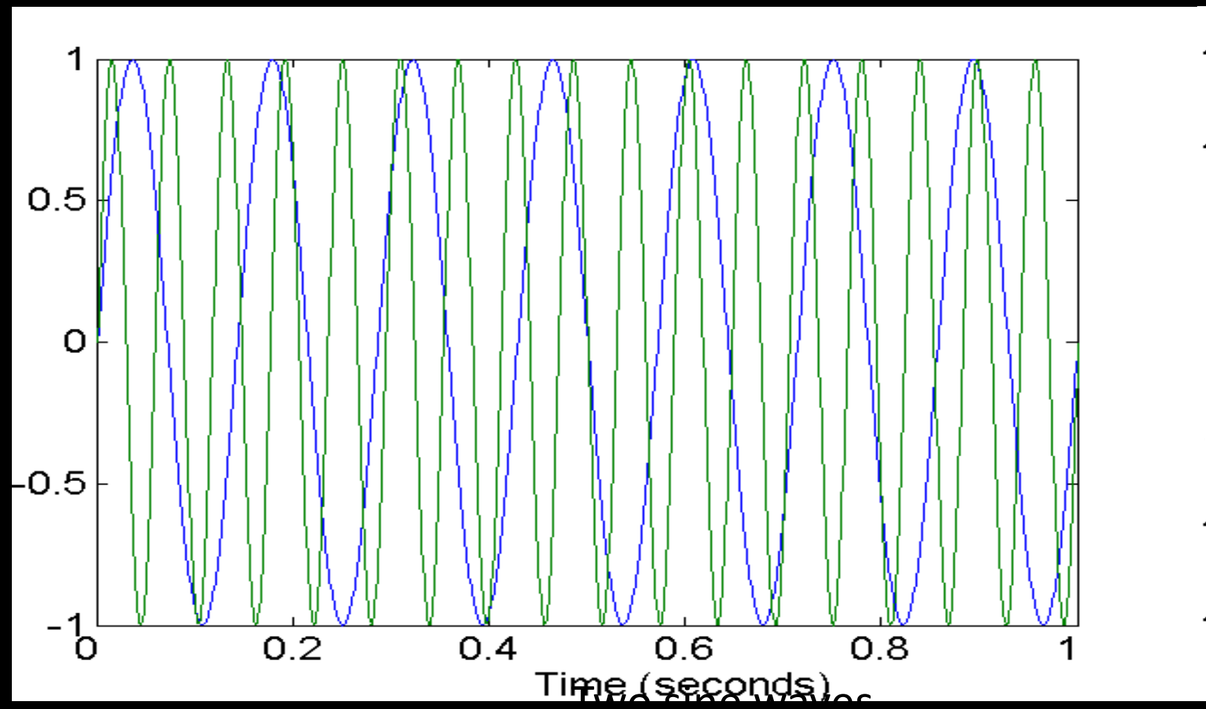
- **What kinds of data quality problems?**
- **How can we detect problems with the data?**
- **What can we do about these problems?**

- **Examples of data quality problems:**
  - noise and outliers
  - missing values
  - duplicate data

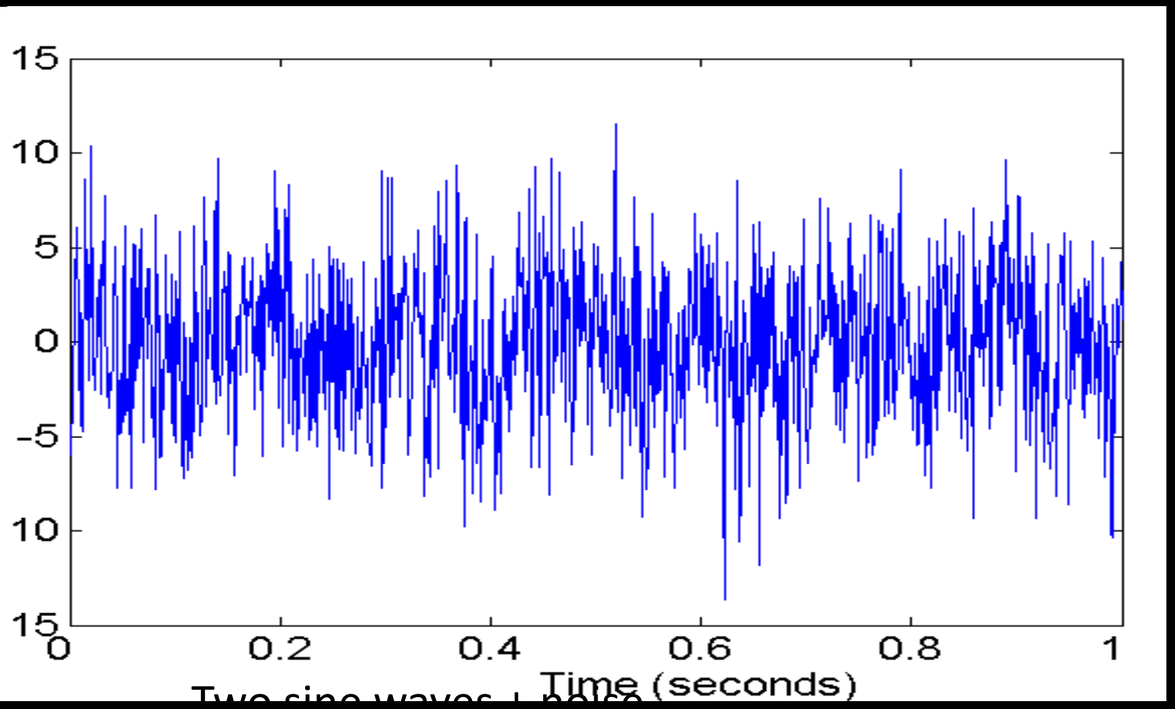
# Noise




- **Noise refers to random modification of original values**
- **Examples:**
  - distortion of a person's voice when talking on a poor phone
  - "snow" on television screen




Two sine waves




Two sine waves + noise

# Outliers

- **Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set**

Fig 1

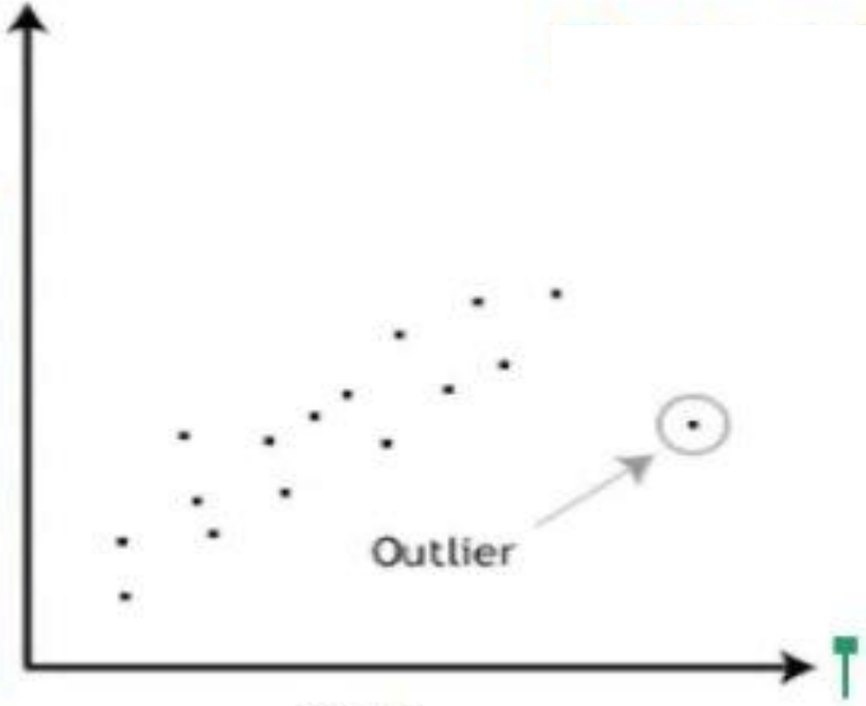



Fig 2

# Missing values

- **Reasons for missing values**
  - Information is not collected (e.g., people decline to give their age and weight)
  - Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)

- **Handling missing values**
  - Eliminate data objects
  - Estimate missing values (imputation)
  - Ignore the missing value during analysis
  - Replace with all possible values (weighted by their probabilities)

# Duplicate data

- **Data set may include data objects that are duplicates, or almost duplicates of one another**
  - Major issue when merging data from heterogeous sources
- **Example:**
  - Same person with multiple email addresses
- **Data cleaning**
  - Includes process of dealing with duplicate data issues

# The Question I Hate the Most…

- How much data do we need?
- A difficult if not impossible question to answer.
- My usual answer: “more is better”.
  - With the warning: “as long as the quality doesn’t suffer”.
- Another popular answer: “ten times the number of features”.

# Data pre-processing

- Aggregation
- Sampling
- Discretization and binarization
- Attribute transformation
- Feature creation
- Feature selection
  - Choose subset of existing features

# Interview Questions : Data Preprocessing

- What is data preprocessing in Machine Learning?
- Why is data preprocessing important?
- What are the main steps involved in data preprocessing?
- What is data cleaning?
- What are missing values?
- What are noisy data?
- What is data transformation?
- What is data reduction?
- What is feature engineering?
- What is feature selection?

# Interview Questions : Missing Data Handling

- What are different types of missing data (MCAR, MAR, MNAR)?
- How do you detect missing values in a dataset?
- What are common techniques to handle missing data?
- When would you drop rows vs impute values?
- What is mean/median/mode imputation?
- What is forward fill and backward fill?
- What is interpolation?
- How does missing data affect model performance?
- What are advanced imputation techniques (e.g., KNN imputation)?
- What are the risks of improper imputation?

# Interview Questions : Data Cleaning & Outliers

- What are outliers?
- How do you detect outliers (Z-score, IQR)?
- How do you handle outliers?
- What is data smoothing?
- What is binning?
- What is noise reduction?
- What is duplicate data and how do you handle it?
- What is data consistency?
- What is data validation?
- What tools do you use for data cleaning?

# Interview Questions : Feature Scaling & Transformation

- What is feature scaling?
- Why is feature scaling necessary?
- What is normalization?
- What is standardization?
- Difference between normalization and standardization?
- What is Min-Max scaling?
- What is Z-score scaling?
- When should you scale data?
- Which algorithms require feature scaling?
- What happens if you don’t scale data?

# Interview Questions : Encoding Categorical Data & Advanced

- What is categorical data?
- What is label encoding?
- What is one-hot encoding?
- What are the drawbacks of one-hot encoding?
- When would you use label encoding vs one-hot encoding?
- What is data leakage?
- How do you prevent data leakage during preprocessing?
- What is a preprocessing pipeline?
- Why should preprocessing be done after train-test split?
- What is feature engineering with examples?
- How do you handle imbalanced datasets?
- How do you preprocess text data in NLP tasks?

# THANK YOU